

Week 1 – Quantum Computing

Classical systems, Complex numbers, Dirac bra-ket notation

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An example of a deterministic system

Consider the following simple example of a classically deterministic system. Suppose we have a graph on 6 vertices, and some number of pebbles on each vertex, given by the following column vector

$$S_0 = \begin{pmatrix} 6 \\ 2 \\ 1 \\ 5 \\ 3 \\ 10 \end{pmatrix},$$

where $(S_0)_{j,1}$ represents the number of pebbles at vertex j .

Deterministic update rule. Each vertex has exactly one outgoing arrow, so after one time step, *all* pebbles at vertex j move to its unique out-neighbor $f(j)$.

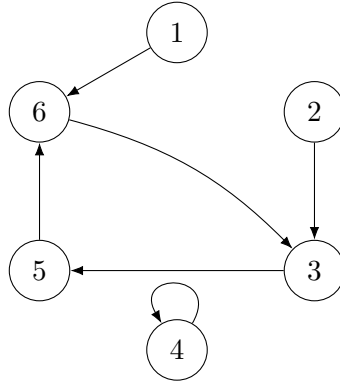


Figure 1: A deterministic directed graph: each vertex has one outgoing edge.

Transition matrix. Define the (column-stochastic) transition matrix M by

$$M_{i,j} = \begin{cases} 1, & \text{if } i = f(j), \\ 0, & \text{otherwise.} \end{cases}$$

For the arrows $1 \rightarrow 6$, $2 \rightarrow 3$, $3 \rightarrow 5$, $4 \rightarrow 4$, $5 \rightarrow 6$, $6 \rightarrow 3$, we get

$$M = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

One time step. The state after one time step is

$$S_1 = MS_0 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 6 \\ 2 \\ 1 \\ 5 \\ 3 \\ 10 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 12 \\ 5 \\ 1 \\ 9 \end{pmatrix}.$$

(For instance, vertex 3 receives all pebbles from vertices 2 and 6, so $(S_1)_3 = (S_0)_2 + (S_0)_6 = 2 + 10 = 12$.)

An example of a classical probabilistic system

We now consider a *probabilistic* system: a single pebble moves randomly on a directed graph. With 3 vertices, we represent the state at time t by a probability vector

$$p_t = \begin{pmatrix} p_t(1) \\ p_t(2) \\ p_t(3) \end{pmatrix}, \quad p_t(i) \geq 0, \quad \sum_{i=1}^3 p_t(i) = 1,$$

where $p_t(i)$ is the probability that the pebble is at vertex i at time t .

A complete directed graph with transition probabilities. Below is a complete digraph (including self-loops) in which each directed edge is labeled by the probability of moving along that edge in one time step.

Transition matrix. Let M be the transition matrix, where $M_{i,j}$ is the probability of moving from vertex j to vertex i in one step. Then

$$M = \begin{pmatrix} 0.5 & 0.25 & 0.25 \\ 0.25 & 0.5 & 0.25 \\ 0.25 & 0.25 & 0.5 \end{pmatrix}.$$

Each column sums to 1 (so M is column-stochastic), and each row also sums to 1, so M is in fact *doubly stochastic*.

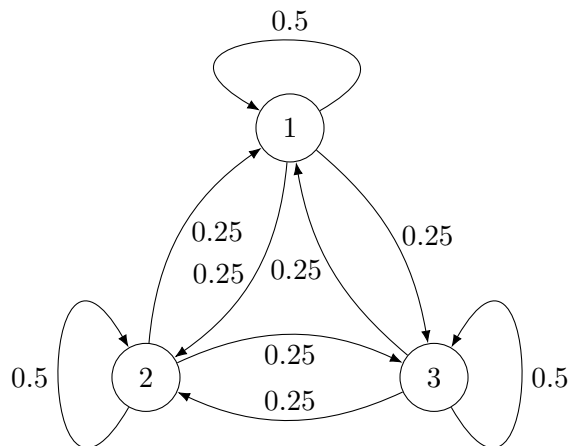


Figure 2: A probabilistic transition rule on a complete digraph (including self-loops).

Initial state and evolution. Suppose the pebble starts at vertex 2 with certainty:

$$p_0 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

After one time step,

$$p_1 = Mp_0 = \begin{pmatrix} 0.5 & 0.25 & 0.25 \\ 0.25 & 0.5 & 0.25 \\ 0.25 & 0.25 & 0.5 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.25 \\ 0.5 \\ 0.25 \end{pmatrix}.$$

After two time steps,

$$p_2 = Mp_1 = \begin{pmatrix} 0.5 & 0.25 & 0.25 \\ 0.25 & 0.5 & 0.25 \\ 0.25 & 0.25 & 0.5 \end{pmatrix} \begin{pmatrix} 0.25 \\ 0.5 \\ 0.25 \end{pmatrix} = \begin{pmatrix} 0.3125 \\ 0.375 \\ 0.3125 \end{pmatrix}.$$

(You can check that the entries of p_1 and p_2 are nonnegative and sum to 1, as required for probability vectors.)

Beyond classical systems: a motivation from experimental physics

It turns out that there exist physical phenomena whose observed behavior cannot be consistently modeled within the classical probabilistic framework described above. A canonical example is the double slit experiment, in which particles such as photons are fired one at a time toward a screen with two slits, and each particle is detected at a single location on a distant screen.

If one attempts to model the experiment classically, one would treat the two slits as providing two mutually exclusive paths to a given detection point. In that case, the probability of detection

with both slits open should equal the sum of the probabilities associated with each slit individually. Experimentally, this is not what is observed: at certain locations, opening the second slit decreases the probability of detection.

Why this occurs is not a question we will attempt to answer. Instead, we take this as evidence that classical probability might sometimes be an inadequate mathematical theory for modeling physical systems. A key step in modeling these types of systems, one that central to quantum mechanics, turns out to be to replace probabilities with *complex amplitudes*. Rather than assigning probabilities to intermediate alternatives and summing them directly, we assign complex numbers whose squared magnitudes give observable probabilities. The amplitudes are first combined linearly, and only then converted into probabilities.

Complex numbers – a minimal review

We conclude Week 1 with a brief review of complex numbers. This section is not meant to be comprehensive; its purpose is to review notation and highlight the geometric interpretation of complex numbers that will be essential in future weeks.

Rectangular form

A complex number z is written as

$$z = a + ib, \quad a, b \in \mathbb{R},$$

where $i = \sqrt{-1}$. The real and imaginary parts of z are denoted by

$$\operatorname{Re}(z) = a, \quad \operatorname{Im}(z) = b.$$

The *complex conjugate* of z is

$$z^* = a - ib.$$

The *magnitude* (or absolute value) of z is defined as

$$|z| = \sqrt{a^2 + b^2}.$$

A basic identity, which we will use frequently, is

$$zz^* = |z|^2.$$

Geometry of complex numbers

We may identify the complex number $z = a + ib$ with the point (a, b) in the plane, called the *Argand plane*. Under this identification, $|z|$ is the Euclidean distance from the origin to (a, b) .

The angle θ that the vector (a, b) makes with the positive real axis is called the *argument* of z , denoted $\arg(z)$. To compute this angle unambiguously, we define the function

$$\theta = \operatorname{atan2}(b, a),$$

where

$$\text{atan2}(b, a) = \begin{cases} \arctan\left(\frac{b}{a}\right), & a > 0, \\ \arctan\left(\frac{b}{a}\right) + \pi, & a < 0, b \geq 0, \\ \arctan\left(\frac{b}{a}\right) - \pi, & a < 0, b < 0, \\ \frac{\pi}{2}, & a = 0, b > 0, \\ -\frac{\pi}{2}, & a = 0, b < 0, \\ \text{undefined} & a = 0, b = 0. \end{cases}$$

Polar form

Using the magnitude $r = |z|$ and angle $\theta = \arg(z)$, any nonzero complex number can be written in *polar form* as

$$z = r(\cos \theta + i \sin \theta).$$

A fundamental identity due to Euler states that

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Using this identity, the polar form of a complex number can be written compactly as

$$z = r e^{i\theta}.$$

In this representation, a complex number is completely specified by two real parameters: **a magnitude r and a phase θ** .

Dirac (bra–ket) notation

Before proceeding further, we introduce a convenient notation for vectors in complex vector spaces that is standard in quantum mechanics.

Kets

A column vector in $\mathbb{C}^n$ will be denoted by a *ket*:

$$|\psi\rangle = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}.$$

The symbol $|\psi\rangle$ is simply notation for a vector. At this stage, you may think of it as an element of $\mathbb{C}^n$.

For example, in $\mathbb{C}^2$ we might write

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C}.$$

Bras

Associated to each ket $|\psi\rangle$ is a *bra*, denoted

$$\langle\psi|.$$

If

$$|\psi\rangle = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix},$$

then the corresponding bra is the conjugate transpose:

$$\langle\psi| = (c_1^* \quad \cdots \quad c_n^*).$$

Thus, passing from $|\psi\rangle$ to $\langle\psi|$ consists of transposing and taking complex conjugates.

Inner product

Given two vectors $|\psi\rangle, |\phi\rangle \in \mathbb{C}^n$, their inner product is written

$$\langle\psi|\phi\rangle = \langle\psi| |\phi\rangle.$$

In coordinates, this is

$$\langle\psi|\phi\rangle = \sum_{j=1}^n \psi_j^* \phi_j.$$

The inner product is skew-symmetric, i.e. $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$.

Norm and normalization

The norm of a ket $|\psi\rangle$ is defined by

$$\| |\psi\rangle \| = \sqrt{\langle\psi|\psi\rangle}.$$

A vector is said to be *normalized* if

$$\langle\psi|\psi\rangle = 1.$$

In quantum mechanics, physical states will be represented by normalized vectors in a complex vector space.